

## LENDING

# The complete loan lifecycle.

From onboarding and credit assessment through issuance, scheduling, recovery, and renewal — how FinFlo runs lending end-to-end, with bank-grade money handling at every step.

DOCUMENT

Loan Management

AUDIENCE

Operations · Lending teams

VERSION

1.0

PRECISION ENGINEERING FOR MODERN FINANCE

## LIFECYCLE AT A GLANCE

## | One continuous flow, from application to closure

Every loan in FinFlo travels the same disciplined path. Each stage hands a clean, auditable state to the next, so balances, repayment ledgers, and reports always reconcile — and the heavy money work is automated by scheduled jobs rather than left to manual reconciliation.



### Onboarding & KYC

The customer is onboarded with CNIC identity capture and document KYC. Sensitive personal fields are encrypted at rest.



### Credit assessment

A credit limit is resolved from the member's share balance and a credit-score band; an optional risk grade and score support the decision.



### Issuance

On approval the loan is created with its principal, rate, duration, EMI, total amount, and start date, and the principal is disbursed.



### Repayment schedule

Installments are tracked through the Repayment ledger; each payment splits into principal and interest portions.



### Recovery

Daily jobs flag overdue installments, accrue late fees, and detect defaults — all in the Asia/Karachi timezone.



### Closure or renewal

A fully repaid loan completes; an eligible customer can roll over, top up, or extend their borrowing through a renewal.

### Why a single flow matters

Because onboarding through payout lives in one system, the loan's principal, accrued interest, late fees, and outstanding balance are never reconstructed from disconnected tools — every number traces back to a ledger row.

## LOAN PRODUCTS &amp; TERMS

## | The fields that define a loan

A FinFlo loan is described by a small, explicit set of terms. **Simple interest** is the default interest type, with EMI-style and compound modes also supported. Money fields are rounded to whole rupees at rest so a fractional rupee can never be persisted.

| FIELD                 | WHAT IT REPRESENTS                                                                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Principal             | The amount lent — the capital the customer receives.                                                                                                               |
| Rate                  | The interest / profit rate applied, as a percentage.                                                                                                               |
| Duration              | The loan term, expressed in months.                                                                                                                                |
| Interest type         | <code>simple</code> (default), <code>emi</code> , or <code>compound</code> .                                                                                       |
| EMI                   | The monthly installment the customer pays.                                                                                                                         |
| Total amount          | Total repayment due over the term — principal plus interest.                                                                                                       |
| Start date            | When the schedule begins; the maturity follows from the duration.                                                                                                  |
| Outstanding principal | Principal still owed, tracked separately from the blended remaining balance so interest accrues on principal only — never on fees or already-capitalized interest. |

#### Illustrative example — repayment schedule

A PKR 120,000 principal over 12 months at a simple rate produces a fixed monthly EMI; each payment is split into an interest portion and a principal portion, and the outstanding principal steps down month by month until the total amount is fully repaid. *Figures are an illustrative example, not an official product or rate.*

## CREDIT LIMITS & ELIGIBILITY

# | Capacity anchored to capital, modulated by behavior

A member's borrowing capacity is computed by the `creditLimitService`. Capacity stays anchored to the member's **share balance** (their business-share investment), and a **credit-score band** raises or lowers it — strong repayment history visibly increases the limit, poor history reduces it. A non-member customer has no share balance, so no automatic limit is enforced for them.

### Share-anchored base

The base limit is derived from the member's share balance, then scaled by their score band (Excellent through Very Poor).

### KYC completeness

Identity and document KYC must be in place before a member can borrow; sensitive fields are stored encrypted.

### Per-tenant scoping

Every limit, member, and loan is scoped to its business owner, so one tenant's lending is fully isolated from another's.

Credit scoring must never break limit resolution — if a score cannot be computed, the service falls back to a neutral band rather than failing.

## REPAYMENT &amp; SCHEDULING

## Every payment is a ledger row

Repayments are recorded as **Repayment** documents linked to the loan, customer, and tenant. Each row carries the total amount paid, its split into `principalAmount` and `interestAmount`, the installment number, and a date. As payments land, the loan's paid amount rises, its remaining balance and outstanding principal fall, and the loan completes when the full total is repaid.

### Installment schedule

Payments are tracked against numbered installments, so the system always knows which periods are settled and which are still due.

### Principal / interest split

Each repayment separates principal from interest, keeping the outstanding-principal figure exact for accrual.

### Partial & early payments

The ledger accepts partial and early payments — the balance moves by exactly what was paid, with no fractional rupee left behind.

### On-time vs late

Whether a payment is on time or late is determined against the installment's due date, which drives downstream overdue and late-fee handling.

Repayment posting is a multi-document money operation, so it runs inside a database transaction and writes an auditable ledger entry — see "Money safety" below.

## LATE FEES

## Automated accrual, never double-charged

Late fees are applied by the `lateFeeService`, driven by a **daily late-fee accrual job** (`runLateFeeAccrual`). A separate **overdue downgrade job** (`runOverdueDowngrade`) moves loans that have missed their due date into the overdue state each day. Both run on a fixed daily schedule in the **Asia/Karachi** timezone.

### One fee per loan per period

A loan records which engine applied its late fee — the daily cron accrual or a manual flat fee. Whichever touches a loan first in a calendar month "owns" it for that month and the other engine skips it, so the two can never stack a fee on the same loan in the same period.

Late-fee accrual is governed by per-tenant settings (whether late fees are enabled, the fee type and rate, and a grace period), so each business applies its own recovery policy.

## DEFAULT DETECTION

## Status transitions, decided by a daily job

A scheduled **default-detection job** (`runLoanDefaultDetection`) runs daily in the Asia/Karachi timezone and advances loans that have fallen far enough behind into the defaulted state, recording when and why. A loan's lifecycle is a small, explicit set of statuses:

| STATUS    | MEANING                                                                                |
|-----------|----------------------------------------------------------------------------------------|
| pending   | Awaiting approval — not yet disbursed.                                                 |
| active    | Disbursed and being repaid on schedule.                                                |
| overdue   | A due installment was missed; flagged by the daily downgrade job.                      |
| defaulted | Sufficiently delinquent; set by the default-detection job with a reason and timestamp. |
| completed | Fully repaid — the loan is closed.                                                     |
| renewed   | Continued through a renewal into a successor loan or extended in place.                |
| rejected  | Declined at approval, with a recorded reason.                                          |

## RENEWALS

## Continuing a customer's borrowing

When a customer is ready to keep borrowing, FinFlo supports three renewal modes, each with explicit linkage between the old and new records so history is never lost.



### Rollover

The existing loan is replaced by a fresh successor loan; the old loan points forward and the new one points back to it.



### Top-up

The customer's borrowing is increased into a new loan that continues from the prior one.



### Extend

No new record is created — the same loan is mutated in place, and its renewal count and last-renewed date are bumped.

Renewal linkage ( `renewedFrom` / `renewedTo` ) keeps the full chain of a customer's borrowing traceable across cycles.

## GROUP LENDING

## Joint-liability groups, individual sub-loans

For the classic microfinance methodology, FinFlo models a standing **LoanGroup** — a joint-liability group of borrowers with a leader (the informational point of contact) and members, plus a guarantee policy. Lending happens through **GroupLoan** cycles: each cycle fans out to **one individual loan per member**, sized to that member's request (per-member allocations).

### The group is the collateral

Under a `joint` guarantee policy, if any member's sub-loan goes overdue or defaults the whole group is flagged `at_risk` and new group lending is frozen until it recovers.

### The engine is unchanged

Each sub-loan is an ordinary loan. The repayment service, late-fee and default jobs, and ledger all run on it as usual — the GroupLoan only ties them together and keeps denormalized roll-ups in sync.

### Aligned schedules

Shared terms (rate, duration, interest type, start date) are applied to every sub-loan so installment dates line up for group-meeting collection.

### Repeat cycles

A group that clears or rolls over one cycle can borrow again as the next cycle, with the same renewal linkage as individual loans.

### Money safety — non-negotiable at every step

Every change to a balance, repayment, distribution, or transfer runs inside a MongoDB session/transaction with idempotent patterns (guarded `findOneAndUpdate` , `$inc` ), and an auditable `FinancialTransaction` ledger row is written for the money movement. `abortTransaction()` is called on every early-return and error path, so a failed operation rolls back cleanly and never leaks an open transaction that would hold locks. All money logic lives in services, not controllers.

